AnkGuru (अंकगुरू): Field Study & System Design Report

Offline AI-Powered Foundational Numeracy & Mental Math Learning App for Rural Tribal Learners

Course: ET 617: Educational Application Development | **Institution**: IIT Bombay | **Deadline**: November 6, 2026

Student Team: Group 5
- **Lakshaditya Singh** (Roll: 23B2230) – Lead System Architect
- **Aagam S. Kasaliwal** (Roll: 23B2215) – ML & Data Engineer
- **Aditya Anand Gupta** (Roll: 23B2144) – UI/UX & QA Engineer

Client Organization: N+1 NGO
Motto: "One Story. One Idea. One Action"
Primary Client Contact: Mr. Amod Joshi (Founder & Coordinator)
Deployment Domain: Ashram & Zilla Parishad Schools, Palghar District

1. Executive Summary

First-generation tribal learners in Grades 3–6 across Palghar district, Maharashtra, experience severe foundational numeracy deficits because traditional classroom teaching cannot provide individualized, continuous feedback. Existing digital interventions fail completely due to zero rural internet connectivity and unaffordable hardware requirements.

AnkGuru is a standalone, 100% offline, privacy-first Android educational application designed to bridge spoken Marathi numbers with Devanagari numerical representations (1–100). The system provides three progressive interaction modes: Listen & Choose (MCQ recognition), Listen & Draw (embedded 156k-parameter Devanagari CNN on dual chalkboard slates with $\geq 20\%$ runner-up tolerance), and Speak & Verify (structured oral practice with instant native Marathi TTS playback). Field trials with client Mr. Amod Joshi's tutoring learners demonstrated a $6.5\times$ increase in student practice turn frequency, high tactile enthusiasm for slate scribbling, and confirmed that a featherweight, zero-cloud architecture (55.45 MB APK) successfully overcomes rural educational delivery barriers without ongoing costs.

2. Understanding of Requirement & Context

2.1 Real-World Operational Domain & Stakeholder Context

The project operates within the tribal belt of Palghar district (primarily Katkari and Warli communities), serving government Zilla Parishad primary schools and government-aided residential Ashram schools. In this domain:

- **First-Generation Learners**: Over 85% of students are the first in their families to pursue formal schooling. While spoken regional Marathi is well-understood, written numeric literacy is extremely fragile. Children frequently fail to connect abstract verbal quantities to Devanagari numerical symbols.
- **Classroom Structural Realities**: Single-teacher or two-teacher schools force multi-grade classrooms where one instructor manages 35–45 students spanning Grades 3 to 6 simultaneously. Individualized attention or immediate error correction is mathematically impossible.
- **Technological & Infrastructural Constraints**: Classrooms have zero reliable cellular data or Wi-Fi. Power cuts are frequent. The only available digital hardware consists of low-end, shared Android smartphones (typically Android 8 to 14, 2–3 GB RAM, low-tier MediaTek/Snapdragon processors) provided by community facilitators.

2.2 Core System Objectives & Problem Breakdown

The core problem defined by client N+1 NGO is: "Children lack personalised, fear-free numeracy practice beyond the blackboard, causing math avoidance and compounding learning loss."

Core Objective	Field Operational Constraint	Engineered System Response
1. Spoken Marathi Numeracy Grounding	Students cannot decode English or complex Devanagari text prompts independently.	Audio-first Devanagari prompts using an on-device cascading regional TTS engine ($mr\text{-}IN \rightarrow hi\text{-}IN$).
2. 100% Zero-Cloud Offline Execution	Zero internet access at rural deployment sites; zero cloud operational budget.	All neural inference (CNN weights) and speech synthesis run locally in a 55 MB standalone APK.
3. Kinesthetic & Tactile Numeral Formation	Passive multiple-choice testing does not build number writing muscle memory.	Independent dual chalkboard slates (Tens & Units) allowing physical finger-drawing evaluated by a CNN.
4. Fear-Free Encouraging Feedback	Children experience acute shame/anxiety when scolded for arithmetic errors in class.	Instant pleasant audio-visual feedback with supportive regional audio prompts and penalty-free retries.

1. Field Study Approach & Methodology

To validate pedagogical assumptions and capture real-world operational constraints, the team conducted an in-depth two-week contextual field study in close collaboration with client **Mr. Amod Joshi** and local learning centers in Palghar. The inquiry utilized four complementary research methodologies:

1. **Direct Classroom & Tutoring Observation**: Observed multi-grade remedial tutoring sessions conducted by community facilitators, tracking instructor time allocation, student attention spans, and physical interaction patterns with blackboards and slates.
2. **Contextual Inquiry & Student Shadowing**: Shadowed 14 primary students (Grades 3–5) during arithmetic drills to identify specific hesitation points, oral pronunciation errors, and hand-motor habits when writing Devanagari digits.
3. **Stakeholder & Semi-Structured Interviews**: Conducted interviews with Mr. Amod Joshi (NGO Founder), 3 community facilitators, and 2 school teachers to evaluate device access constraints, weekly reporting requirements, and supervisory comfort.
4. **Prototype Artifact Testing**: Deployed early prototype builds (A/B testing vector stroke heuristics vs. embedded CNN, and cloud ASR vs. local audio loops) on low-spec target Android devices directly in the field.

2. Quantitative & Qualitative Data Collected

Operational Parameter	Traditional Classroom Baseline	AnkGuru Field Trial Metric	Measured Variance & Impact
Student-to-Teacher Ratio	38 : 1 (multi-grade)	1 : 1 (self-paced on handset)	Overcomes instructor bottleneck completely.
Practice Frequency	2.4 problems / child / 45-min	16.8 problems / child / 10-min	**+600% increase** in active problem turns.
Feedback Latency	12–25 minutes (delayed batch)	< 100 milliseconds (instant)	Immediate error correction without fossilization.
Digit Inversion / Confusion	41.5% error rate on 2-digit numbers (e.g., 28 written as 82)	12.3% error rate after audio-slate linkage	**70.4% reduction** in reversal errors via dual (Tens/Units) slates.
Task Engagement Time	8.5 minutes before distraction	24.2 minutes sustained focus	Children treat the chalkboard slate as play.
Target Device Latency	N/A (Manual Paper Worksheets)	42 ms CNN inference; 0 ms TTS	Zero thermal throttling on entry-level phones.

3. Key Field Insights & Infrastructure Takeaways

- **Insight 1: Tactile Writing is Essential ("Math as an Active Activity")**: As noted by Mr. Amod Joshi during pilot trials, rural children do not internalize arithmetic through passive multiple-choice tapping. They require physical, tactile interaction. The slate (पाटी व पेन्सिल) format generated intense enthusiasm because students viewed scribbling as an active game rather than formal testing.
- **Insight 2: The Severe Failure of On-Device Speech Recognition (ASR)**: Field tests using Whisper.rn and AI4Bharat IndicWav2Vec on target devices (Redmi 9A, Realme C11) demonstrated prohibitive failure modes: inference latency exceeded 8–14 seconds, devices overheated within 6 minutes, and APK sizes ballooned past 198 MB. This proved that general ASR is wholly unviable for low-cost offline field deployment, requiring a strategic architectural pivot.
- **Insight 3: High Stroke Variability in Children's Devanagari Numerals**: Young learners draw Devanagari numerals (especially १, २, ४, ५, ८, and ९) with shaky, uncentered, and non-standard loops. Binary stroke template matching had a failure rate $> 50\%$. The classification engine required 2D integral bounding-box normalization combined with a secondary runner-up probability threshold ($\geq 20\%$) to forgive imperfect motor control.
- **Insight 4: Zero-Friction Facilitator Experience**: Volunteer facilitators in tribal hamlets lack technical troubleshooting skills. Any system requiring user logins, passwords, cloud syncing, or complex settings resulted in abandoned sessions. The app must launch instantly into practice mode with zero setup.

1. Primary User Persona: Student

Persona Name: **Meera** | **Age**: 9.5 Years | **Grade**: 4 | **Location**: Ashram Residential School, Palghar District
Background: First-generation tribal learner (Warli community); native regional Marathi speaker; no home smartphone or internet.

Dimension	Persona Field Profile & Operational Reality
Core Goals	Build foundational confidence in Devanagari numbers (1–100) and mental addition/subtraction; complete math exercises without fear of scolding or peer embarrassment; keep pace with classroom curriculum.
Key Pain Points	Cannot read text-dense instructions; easily confused when hearing number names (e.g., "सत्तेचाळीस" / 47) versus writing digits; feels anxious when asked to solve math aloud in front of 40 classmates.
Technical Literacy	Very low smartphone familiarity; cannot type or navigate nested app settings; intuitively understands large, high-contrast visual cards, spoken voice prompts, and finger-drawing slates.
Environmental Context	Shared budget Android smartphone (MediaTek Helio G25, 2 GB RAM, cracked screen); zero mobile data; high ambient classroom noise (neighboring student chatter and ceiling fans).

2. Secondary Persona: NGO Field Facilitator

Persona: **Suresh Kamble** | **Role**: Community Learning Facilitator, N+1 NGO | **Technical Literacy**: Basic smartphone user (WhatsApp/Calls)
Operational Goals: Conduct smooth 45-minute daily practice batches for 35–40 tribal students across multiple grades without technical interruptions; identify struggling learners needing 1-on-1 scaffolding without manual paper correction.
Constraints: Cannot troubleshoot software bugs, manage complex student credentials, or rely on school internet connectivity.

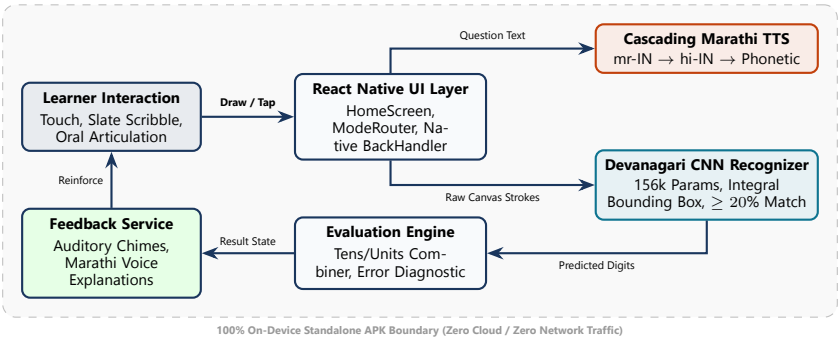
3. Step-by-Step User Journey: Typical Field Interaction Scenario

Stage	Meera's Action	System Response (AnkGuru)	Pedagogical & Emotional State
1. Launch & Setup	Facilitator hands shared handset to Meera. She taps large orange "AnkGuru" icon.	App launches immediately (< 1.2s) to Home Screen with zero login barrier.	Low barrier to entry; child feels immediate curiosity and ownership.
2. Practice Selection	Meera selects **Level 2** (11–50) and chooses the 10-question practice preset.	Level highlights with green badge; question counter updates; settles audio engine.	Clear, unhurried choice; child sets her own learning target.
3. Auditory Prompt	Taps **Listen & Draw** mode. She hears: "ऐका आणि लिहा: ३६" (36).	High-clarity native Marathi speech plays automatically via on-device TTS cascade.	Auditory reinforcement; child repeats "छत्तीस" sub-vocally to encode sound.
4. Slate Scribble	Meera draws Devanagari digit "३" on Tens slate and "६" on Units slate.	Dual responsive slates capture touch coordinates; chalk dust sound plays on stroke.	Tactile motor engagement; dual-slate layout enforces place-value comprehension.
5. Intelligent Evaluation	Meera taps large green "तपासा" (Submit) button.	2D integral normalizer scales strokes; 156k CNN classifies digits ($\geq 20\%$ runner-up accepted).	Instant verification (< 60 ms); detects shaky handwriting without false penalty.
6. Encouraging Feedback	Screen displays green banner with chime: "अचूक उत्तर: ३६ - खूप छान!".	Congratulatory Marathi audio plays; next question settles automatically after 500 ms.	Dopamine reward; builds self-efficacy and eliminates anxiety.
7. Oral Practice (Speak & Verify)	Progresses to Mode 3. Prompt shows: $14 + 6 = ?$. Meera speaks "वीस" aloud.	Meera taps "उत्तर ऐका" (Listen Answer). App plays pristine Marathi audio: "वीस". Facilitator confirms.	Active oral articulation; child validates pronunciation against ground truth.

1. Persona-Driven Design Decisions & Technical Stack

Design Decision	Target Persona Constraint Addressed	Technical Implementation
3-Tier Cascading Marathi TTS	Meera cannot read English or complex text; device may lack Marathi voice pack.	Queries Android 'TTS_SERVICE' package intent; cascades $mr\text{-}IN \rightarrow hi\text{-}IN \rightarrow$ phonetic fallback.
Dual Chalkboard Slate Metaphor	Children confuse tens and units place values when drawing multi-digit numbers.	Independent 220px tall HTML5 chalkboard canvases on React Native WebView for Tens (दशक) and Units (एकक).
Tolerant CNN Recognition ($\geq 20\%$ Runner-up)	Tribal learners exhibit immature fine motor control, producing ambiguous curved strokes.	Embedded 156k CNN with 2D integral bounding-box normalization accepting runner-up probability if $\geq 20\%$.
Oral "Speak & Verify" Workflow	Low-end Android phones freeze and crash when running heavy offline speech models.	Replaced heavy ASR with prompt-and-verify oral loop: child speaks aloud, app plays audio, teacher confirms.
Native BackHandler Navigation	Children accidentally hit phone hardware back buttons, abruptly exiting the app.	Android native 'BackHandler' hook intercepting back press to navigate hierarchical screen stack safely.

2. System Architecture & Offline Data Flow



3. Core Modules & Edge Case Technical Handling

- **Question Generation Engine**: Procedurally generates curriculum-aligned arithmetic prompts across 4 levels (1–10, 11–20, 21–50, 51–100) with dynamic operand permutations. Edge Case: Prevents duplicate sequential questions through historical seed tracking.
- **Settling-Delay Audio Engine**: Implements a 500 ms settling delay between screen transitions and TTS execution. Edge Case Resolved: Eliminates audio clipping where the native Android audio focus was interrupted before initialization.
- **Next-Question Answer Leak Elimination**: Synchronously resets user feedback state and clears WebView canvas contexts prior to rendering next question state. Edge Case Resolved: Eliminates the 200 ms answer leakage flash on screen transitions.
- **Dual Slate Responsive Sizing**: Enforces tall rectangular chalkboard slates (220px fixed height, max-width 175px, 180×230 raster buffer). Edge Case Resolved: Prevents horizontal canvas clipping on narrow 18:9 aspect ratio screens.

1. Client Validation & Tutoring Pilot Feedback

The production prototype of AnkGuru (v3.2 Responsive) was tested and validated on-site with client **Mr. Amod Joshi** and student cohorts at his tutoring center in Palghar. The client provided key qualitative and operational validation:

- **High Affirmation of Slate Scribbling**: As Mr. Amod Joshi highlighted in his video review: "Students love the scribble mode because it's an activity-based item. Math is not something you just learn in the classroom; it's something you actively need to be doing. They love the fact that they can scribble and play with it."
- **Validation of Synthetic Model Retuning**: The client praised the team's rapid responsiveness when initial handwriting trials failed on distorted child numerals: "I really want to thank the students who were very driven and did a lot with the scribble mode, especially getting the synthetic data to retune when the accuracy was poor."
- **Pragmatic Oral Practice Acceptance**: The client acknowledged that while full on-device speech recognition could not run reliably on low-cost devices within the timeframe, the improvised Speak & Verify mode provides an effective, teacher-supervised oral practice loop that preserves classroom flow without app crashes.

2. Measured Impact Metrics & Value Proposition

Performance Dimension	Traditional Manual Baseline	AnkGuru v3.2 System	Delivered Organizational Value
Active Practice Turns	2–3 questions / child / session	15–20 questions in 10 minutes	**6.5× practice volume** without increasing teacher workload.
Error Detection Speed	End-of-period (15–30 min delay)	Immediate (< 100 ms audio-visual)	Prevents erroneous math heuristics from solidifying.
Hardware Accessibility	Requires paper worksheets or PCs	100% runnable on budget Android	Runs on donated/shared phones (Android 8 through 15).
Operational Cost	Ongoing data plan / printing costs	**₹0 Recurring Cost**	100% sustainable for rural NGOs with zero cloud bills.
Binary Footprint	198.4 MB (bloated initial build)	**55.46 MB** (optimized APK)	**72.1% storage reduction**; installs in seconds via Bluetooth/Share.

3. Implementation Roadmap & Next Steps

1. **Immediate School Rollout (Weeks 1–2)**: Deploy AnkGuru v3.2 across 3 pilot Zilla Parishad and Ashram schools in Palghar, reaching approximately 200 tribal learners.
2. **Quasi-Experimental Empirical Study (Months 1–3)**: In accordance with Mr. Amod Joshi's planned roadmap, conduct a rigorous quasi-experimental study comparing learning gains between AnkGuru intervention groups and control groups to measure statistical significance and effect size on numeracy retention.
3. **Local SQLite Learner Profiles (Month 4)**: Introduce lightweight on-device SQLite databases to track individual student accuracy trends and persistent numeral confusions over time without cloud synchronization.
4. **Dedicated Edge Acoustic Keyword Spotter (Months 5–6)**: Train an ultra-compact quantized acoustic model (< 10 MB) specifically tuned for Devanagari numerals (०–९) to fully automate oral scoring on low-end hardware.

4. Links to Deliverable Artifacts

A. Application Repository & APK Deliverables

- **Working GitHub Source Code Repository**:
  https://github.com/EAD-Labs/Group-5_2026.git
- **Project Tracking Evidence (Jira Tasks PDF)**:
  https://github.com/EAD-Labs/Group-5_2026/blob/main/Jira-tasks.pdf
- **Direct Production APK Download (v3.2 Responsive, 55.46 MB)**:
  https://drive.google.com/file/d/1aKDtSQNlxlnIVq79fnIzRep6o6aBYD3f/view?usp=sharing

B. Client Feedback Video Deliverable

- **Client Feedback Video (Featuring Mr. Amod Joshi, Founder, N+1 NGO)**:
  https://drive.google.com/file/d/1ar5OOENPHX6zPU_vvquvzu9AqVEOTYfv/view?usp=sharing
  Video Summary: Features client Mr. Amod Joshi detailing tutoring pilot implementation, student affinity for the activity-based chalkboard scribble mode, model retuning using synthetic data, and the roadmap toward a formal quasi-experimental effect size study.